MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

DIPLOMA IN LIBRARY AND INFORMATION SCIENCE

BICT 1103: COMPUTER COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

DATE: MARCH, 2020 **TIME ALLOWED: 2 HOURS 30 MINUTES**

INSTRUCTIONS

1. This paper has **SIX** questions.
2. Answer **ALL** questions.
3. Marks for each question are indicated.
4. Write your answers clearly.
5. Indicate the question number for each answer.
6. This paper contains **TWO** pages. **Please check**

1. Differentiate the following terms:
   a. Website and Internet. **[2 marks]**
   b. Server and Database. **[2 marks]**
   c. Digital signal and Analogue signal. **[2 marks]**
   d. Operating System and Utility programs. **[2 marks]**
   e. Bandwidth and Data rate. **[2 marks]**

2.
   a. Explain how a laser printer works. **[3 marks]**
   b. Describe the components of a Central Processing Unit. **[6 marks]**
   c. Describe the first **three** generations of computers. **[6 marks]**
   d. State any **six** components of microcomputer system unit. **[6 marks]**

3.
   a. Differentiate between an octal number system and a hexadecimal number system. **[4 marks]**
   b. Convert the following numbers:
      i. 11101111_2 to denary. **[4 marks]**
      ii. A46F to binary. **[5 marks]**
   c. Compute the following:
      i. $73546_8 - 67674_8$ Using complement. **[5 marks]**
      ii. EB6DF ÷ B7CAD. **[5 marks]**

4.
   a. Describe three technologies that are used for storing data. **[6 marks]**
   b. Explain the meanings of the following terms as they are used in data storage;
      i. DVD +-RW. **[2 marks]**
      ii. CD-R. **[2 marks]**
      iii. Flash Memory. **[2 marks]**
      iv. Removable storage **[2 marks]**

5.

a. Explain **six** roles that are carried out by Operating system. **[12 marks]**

b. Describe **three** system software tools you can find Windows operating systems. **[9 marks]**

6. Explain any **five** ways in which ICT can benefit the society. **[10 marks]**

End of question paper